INTRODUCTION TO ANALYSIS

1.1 ANALYSIS

Mathematical analysis is the branch of mathematics dealing with limits and related theories, such as differentiation, integration, measure, infinite series, and analytic functions. Mathematical analysis formally developed in the 17th century during the scientific revolution, but many of its ideas can be traced back to earlier mathematicians. Early results in analysis were implicitly present in the early days of ancient Greek mathematics.
These theories are usually studied in the context of real and complex numbers and functions. Analysis evolved from calculus, which involves the elementary concepts and techniques of analysis.

- Real analysis

Real analysis (traditionally, the theory of functions of a real variable) is a branch of mathematical analysis dealing with the real numbers and real-valued functions of a real variable. In particular, it deals with the analytic properties of real functions and sequences, including convergence and limits of sequences of real numbers, the calculus of the real numbers, and continuity, smoothness and related properties of real-valued functions.

- Complex analysis

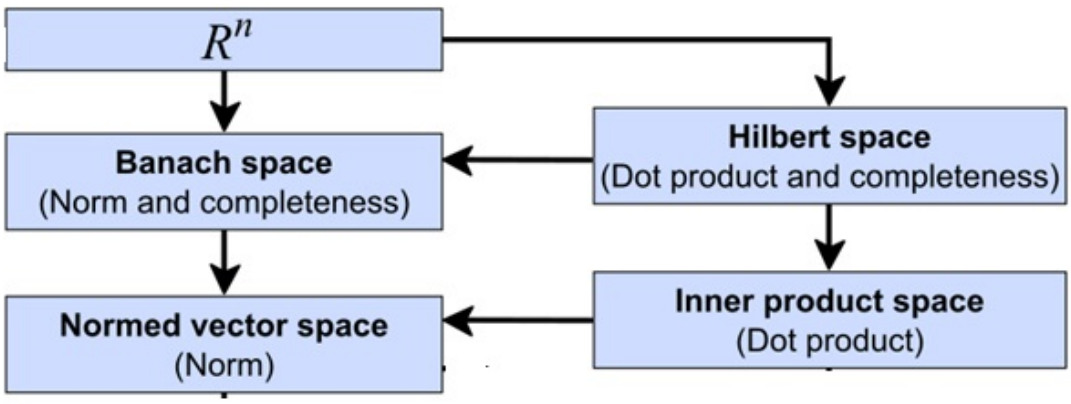
Complex analysis, traditionally known as the theory of functions of a complex variable, is the branch of mathematical analysis that investigates functions of complex numbers. It is useful in many branches of mathematics, including algebraic geometry, number theory, applied mathematics; as well as in physics, including hydrodynamics, thermodynamics, mechanical engineering, electrical engineering, and particularly, quantum field theory.

Complex analysis is particularly concerned with the analytic functions of complex variables (or, more generally,meromorphic functions). Because the separate real and imaginary parts of any analytic function must satisfy Laplace's equation, complex analysis is widely applicable to two-dimensional problems in physics.

- Functional analysis

Functional analysis is a branch of mathematical analysis, the core of which is formed by the study of vector spacesendowed with some kind of limit-related structure (e.g. inner product, norm, topology, etc.) and the linear operators acting upon these spaces and respecting these structures in a suitable sense. The historical roots of functional analysis lie in the study of spaces of functions and the formulation of properties of transformations of functions such as the Fourier transform as transformations defining continuous, unitary etc. operators between function spaces. This point of view turned out to be particularly useful for the study of differential and integral equations.

Functional Analysis and its Applications is devoted to current problems of functional analysis - representation theory, theory of abstract and functional spaces, theory of operators, spectral theory, theory of operator equations, and the theory of normed rings. Functional analysis is a methodology that is used to explain the workings of a complex system. The basic idea is that the system is viewed as computing a function(or, more generally, as solving an information processing problem).



1.2 DIFEERENCE BETWEEN REAL,COMPLEX AND FUNCTIONAL ANALYSIS

Real Analysis is the study of the real numbers and functions of a real variable, including aspects of limits, continuity, infinite series, differentiation and integration.

Complex Analysis is the study of the complex numbers and functions of a complex variable, including aspects of limits, continuity, infinite series, differentiation and integration.

Functional Analysis is the study of spaces of functions and operators on those spaces, and more abstractly the study of infinite-dimensional vector spaces endowed with a norm or an inner product (or just a topology).

1.3 THE APPLICATION OF ANALYSIS IN MATHEMATICAL AND THEORETICAL PHYSICS

Below, those branches of mathematical physics are given in which some part of functional analysis is applied.

1. The spectral theory of operators is applied in all theories of quantum physics: in quantum -body theory, in quantum field theory and in quantum statistical mechanics. In addition, spectral theory is applied in the study of models of dynamical systems in classical mechanics, in the study of linearized equations in hydrodynamics, in the study of Gibbs fields, etc.

2. Scattering theory is applied in quantum physics. It should be noted that the modern mathematical theory of scattering arose first of all in physics. In recent years scattering theory (the inverse problem) has often been applied in integrating non-linear model equations in mathematical physics.

3. Banach algebras are applied in quantum field theory, especially in so-called axiomatic field theory, and in studying various integrable models of a quantum field and statistical mechanics. Von Neumann algebras are also used in these questions.

4. Perturbation theory, mainly the perturbation theory for linear operators, is applied in almost-all domains of mathematical physics: in quantum field theory

and in statistical mechanics, both equilibrium and non-equilibrium (especially in investigating the so-called kinetic equations, the compound spectra of multi-particle systems, etc.).

5. Functional integration and measures in function spaces are applied in constructive quantum field theory and in quantum statistical mechanics.

6. Various integral representations (Riesz' theorem, the Krein-Mil'man theorem, Choquet's theorem, and others) are applied in axiomatic quantum field theory and in statistical mechanics.

7. Vector spaces (mainly Hilbert spaces) are applied in quantum theory and in statistical physics.

8. Generalized functions are applied everywhere in mathematical physics as an important analytic tool (see also Generalized function).

1.4 OPERATOR THEORY

In mathematics, operator theory is the branch of functional analysis that focuses on bounded linear operators and also includes closed operators and nonlinear operators. Spectral theory or spectral analysis of linear operators is one of the most dynamic area in operator theory. The study, which depends heavily on the topology of function spaces, is a branch of functional analysis.

1.4.1 Single operator theory

Single operator theory deals with the properties and classification of operators, considered one at a time. For example, the classification of normal operators in terms of their spectra falls into this category.

- Spectrum of operators

The spectral theorem is any of a number of results about linear operators or about matrices. In broad terms the spectral theorem provides conditions under which an operatoror a matrix can be diagonalized (that is, represented as a diagonal matrix in some basis). This concept of diagonalization is relatively straightforward for operators on finite-dimensional spaces, but requires some modification for operators on infinite-dimensional spaces. In general, the spectral theorem identifies a class of linear operators that can be modelled by multiplication operators, which are as simple as one can hope to find. In more abstract language, the spectral theorem is a statement about commutative C*-algebras. See also spectral theory for a historical perspective.

Examples of operators to which the spectral theorem applies are self-adjoint operators or more generally normal operators on Hilbert spaces.

The spectral theorem also provides a canonical decomposition, called the spectral decomposition, eigenvalue decomposition, or eigendecomposition, of the underlying vector space on which the operator acts.

- Normal operators

A normal operator on a complex Hilbert space H is a continuous linear operator N : $H \to H$ that commutes with its hermitian adjoint N^*, that is: $NN^* = N^*N$.

Normal operators are important because the spectral theorem holds for them. Today, the class of normal operators is well understood. Examples of normal operators are

$\Rightarrow$ unitary operators:$N^* = N^{-1}$
$\Rightarrow$ Hermitian operators (i.e., selfadjoint operators): $N^* = N$; (also, anti-selfadjoint operators:$N^* = -N$)
$\Rightarrow$ positive operators: $N = MM^*$
$\Rightarrow$ normal matrices can be seen as normal operators if one takes the Hilbert space to beC_n

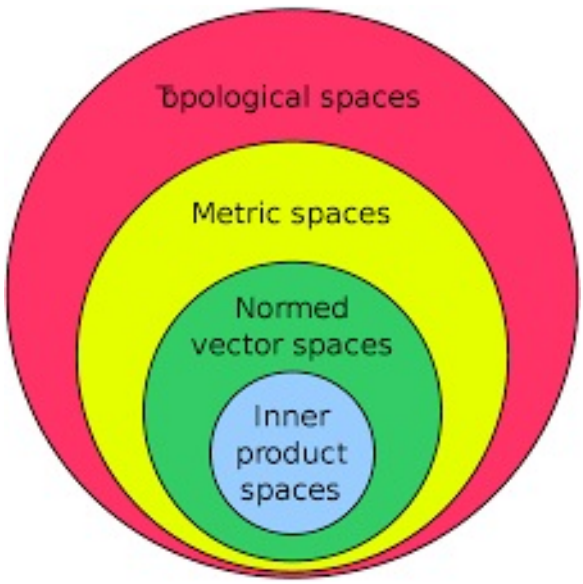
The spectral theorem extends to a more general class of matrices. Let A be an operator on a finite-dimensional inner product space. A is said to be normal if $A^*A = AA^*$. One can show that A is normal if and only if it is unitarily diagonalizable: By the Schur decomposition, we have$A = UTU^*$, where U is unitary and T upper-triangular. Since A is normal, $TT^* = T^*T$. Therefore, T must be diagonal since normal upper triangular matrices are diagonal. The converse is obvious.

In other words, A is normal if and only if there exists a unitary matrix U such that where D is a diagonal matrix. Then, the entries of the diagonal of D are the eigenvalues of A. The column vectors of U are the eigenvectors of A and they are orthonormal. Unlike the Hermitian case, the entries of D need not be real.

PRELIMINARIES

To introduce Hilbert spaces, and more precisely the Hilbert spaces on the field of complex numbers, which represent the abstract environment in which Quantum Mechanics is developed.

To arrive at Hilbert spaces, we proceed gradually, beginning with spaces mathematically less structured, to move toward more and more structured ones, considering, in order of complexity:



linear or vector spaces, in which the points of the space are called vectors, and the operations are the sum between two vectors and the multiplication by a scalar;

normed vector spaces, in which the concept of norm of a vector x is introduced, indicated by$\|x\|$, from which one can obtain the distance between two vectors x and y as d $(x, y) = \|x - y\|$;

vector spaces with inner product, in which the concept of inner-product between two vectors x, y is introduced, and indicated in the form (x, y), from which the norm can be obtained as $\|x\|$=$\langle x, x\rangle^{1/2}$, and then also the distance d (x, y);

Hilbert spaces, which are vector spaces with inner product, with the additional property of completeness.

We will start from vector spaces, then we will move on directly to vector spaces with inner product and, eventually, to Hilbert spaces. For vectors, we will initially adopt the standard notation $(x, y, etc.)$, and subsequently we will switch to Dirac's notation, which has the form x, y, etc., universally used in Quantum Mechanics.

2.1 DEFINTION OF VECTOR SPACE

A vector space (also called a linear space) is a collection of objects called vectors, which maybe added together and multiplied ('scaled') by numbers, called scalars. Scalars are often taken to be real numbers, but there are also vector spaces with scalar multiplication by complex numbers, rational numbers, or generally any field. The operations of vector addition and scalar multiplication must satisfy certain requirements, called axioms

Vector addition and scalar multiplication: a vector v (blue) is added to another vector w (red, upper illustration). Below, w is stretched by a factor of 2, yielding the sum v + 2w.

A vector space on a field F is essentially an Abelian group, and therefore a set provided with the addition operation + , but completed with the operation of multiplication by a scalar belonging to F.

Here we give the definition of vector space in the field of complex numbers , as it is of interest to Quantum Mechanics.

A vector space in the field of complex numbers C is a nonempty set V, whose elements are called vectors, for which two operations are defined. The first operation, addition, is indicated by + and assigns to each pair $(x, y) \in VV$ a vector $x + y \in V$. The second operation, called multiplication by a scalar or simply scalar multiplication, assigns to each pair $(a, x) \in CV$ a vector $ax \in V$. These operations must satisfy the following properties, for $x, y, z \in V$ and $a, b \in C$:

1. $x + (y + z) = (x + y) + z$ (associative property),
2. $x + y = y + x$ (commutative property),
3. V contains an identity element 0 with the property $0 + x = x$, $\forall x \in V$,
4. V contains the opposite (or inverse) vector $-x$ such that $-x + x = 0$, $\forall x \in V$,
5. $a(x + y) = ax + ay$,
6. $(a + b)x = ax + bx$.

Notice that the first four properties assure that V is an Abelian group or commutative group, and, globally, the properties make sure that every linear combination

$$a1x1 + a2x2 + \ + a_n x_n \ , \ a_i \in C, \ x_i \in V \text{ is also a vector of V.}$$

A normed linear space is a linear space N in which to each vector x, there corresponds a real number denoted by satisfying

1. $\|x\| \geq 0$
2. $\|x\| = 0$ if and only if $x = 0$
3. $\|x + y\| \leq \|x\| + \|y\| \, 0$ for every $x, y \in N$
4. $\|\alpha x\| \leq |\alpha| \, \|x\|$, where α is any scalar.

A normed linear space is said to be complete if every Cauchy sequence has a limit in X , that is, if$\|x_m - x_n\| \to 0$ as $n \to \infty$ there exists $x_0 \in X$ such that $\|x_n - x_0\| \to 0$ as $n \to \infty$.

A Banach space is a complete normed linear space.

2.2 HIBERT SPACES

In this section we take a closer look at linear continuous maps between Hilbert spaces. These are often called bounded operators, and the branch of Functional Analysis that studies these objects is called 'Operator Theory'.The standard notations in Operator Theory are as follows. The mathematical concept of a Hilbert space, named after David Hilbert, generalizes the notion of Euclidean space. It extends the methods of vector algebraand calculus from the two-dimensional Euclidean plane and three-dimensional space to spaces with any finite or infinite number of dimensions. A Hilbert space is an abstract vector space possessing the structure of an inner product that allows length and angle to be measured.

Furthermore, Hilbert spaces are complete: there are enough limits in the space to allow the techniques of calculus to be used.

Hilbert spaces arise naturally and frequently in mathematics and physics, typically as infinite-dimensional function spaces. The earliest Hilbert spaces were studied from this point of view in the first decade of the 20th century by David Hilbert,Erhard Schmidt, and Frigyes Riesz. The success of Hilbert space methods ushered in a very fruitful era for functional analysis.

- Inner Product space

Let X be a vector space over the complex scalars C , there exists a complex number $\langle x, y\rangle$ for each pair of vectors $x, y \in X$ satisfying the following properties, then $\langle x, y\rangle$ is called inner product of x and y.

1. $\langle x, x \rangle \geq 0$ for all x in X and $\langle x, x \rangle = 0$ if and only if $x = 0$
2. $\overline{\langle x, y \rangle} = \langle y, x \rangle$ for all x and y in X
3. $\langle x + y, z \rangle = \langle x, z \rangle + \langle z, y \rangle$ for all $x, y, z \in X$
4. $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$ for all x and y in X and all complex numberλ

A complex vector space X having the inner product is said to be inner product space.

- Definition of Hilbert space

A Hilbert space is a complete inner-product vector space.

It must be observed that a finite dimensional vector space is always complete, as it is closed with respect to all its sequences, and therefore it is always a Hilbert space. Instead, if the space is infinite dimensional, the completeness is not ensured, and therefore it must be added as a hypothesis, in order for the inner-product vector space to become a Hilbert space. At this point, we want to reassure the reader: the theory of optical quantum com- munications will be developed at a level that will not fully require the concept of a Hilbert space, but the concept of inner-product vector space will suffice. Nonethe- less, the introduction of the Hilbert space is still done here for consistency with the Quantum Mechanics literature. From now on, we will assume to operate on a Hilbert space, but, for what we just said, we can refer to an inner-product vector space. Operators on Hilbert spaces

2.3 LINEAR OPERATORS

The continuous linear operators $A : H_1 \to H_2$ from a Hilbert spaceH_1 to a second Hilbert space H_2 are bounded in the sense that they map bounded sets to bounded sets. Conversely, if an operator is bounded, then it is continuous. The space of such bounded linear operators has a norm, the operator norm given by

$$\|A\| = \{\|Ax\| : \|x\| \leq 1\}$$

The sum and the composite of two bounded linear operators is again bounded and linear. For y in H_2, the map that sends $x \in H_1$ to $\langle Ax, y\rangle$ is linear and continuous, and according to the Riesz representation theorem can therefore be represented in the form

$$\langle x, A^*y\rangle = \langle Ax, y\rangle$$

for some vector A^*y in H_1. This defines another bounded linear operator $A^* : H_2 \to H_1$, the adjoint of A. One can see that $A^{**} = A$. The set B(H) of all bounded linear operators on H (operators $H \to H$), together with the addition and composition operations, the norm and the adjoint operation, is a C^*-algebra, which is a type of operator algebra.

An element A of B(H) is called 'self-adjoint' or 'Hermitian' if $A^* = A$. If A is Hermitian and $\langle Ax, x\rangle \geq 0$ for every x, then A is called 'nonnegative', written $A \geq 0$; if equality holds only when $x = 0$, then A is called 'positive'. The set of self adjoint operators admits a partial order, in which $A \geq B$if $A - B \geq 0$. If A has the form B^*B

for some B, then A is nonnegative; if B is invertible, then A is positive. A converse is also true in the sense that, for a non-negative operator A, there exists a unique non-negative square root B such that

$$A = B^2 = B^*B$$

In a sense made precise by the spectral theorem, self-adjoint operators can usefully be thought of as operators that are 'real'. An element A of B(H) is called normal if $A^*A = AA^*$.

Normal operators decompose into the sum of a self-adjoint operators and an imaginary multiple of a self adjoint operator

$$A = \frac{A+A^*}{2} + i\frac{A-A^*}{2i}$$

that commute with each other. Normal operators can also usefully be thought of in terms of their real and imaginary parts.

An element U of B(H) is called unitary if U is invertible and its inverse is given by U^*. This can also be expressed by requiring that U be onto and $\langle Ux, Uy\rangle = \langle x, y\rangle$for all$x, y \in H$. The unitary operators form a group under composition, which is the isometry group of H. An element of B(H) is compact if it sends bounded sets to relatively compact sets. Equivalently, a bounded operator T is compact if, for any bounded sequence x_k, the sequence Tx_k has a convergent subsequence. Many integral operators are compact, and in fact define a special class of operators known as Hilbert-Schmidt operators that are especially important in the study of integral

equations. Fredholm operators differ from a compact operator by a multiple of the identity, and are equivalently characterized as operators with a finite dimensional kernel and cokernel. The index of a Fredholm operator T is defined by

$$indexT = dimkerT - dumcokerT$$

A linear operator T on a Hilbert space H is said to be bounded, if there exists $c > 0$ such that $\|Tx\| \leq c\,\|x\|$ for all $x \in H$ is defined by

$$\|T\| = inf\,\{c > 0 : \|Tx\| \leq c\,\|x\|\}$$

If T is an operator on a Hilbert space H, then there exist a unique operator T^* on H such that $\langle Tx, y\rangle = \langle x, T^*y\rangle$for all $x, y \in H$

This operator T^* is called an adjoint of T.

If T is an operator defined on a Hilbert space H then the linear operator T^* on H^* defined by $T^*(f) = f \circ T$ for every $f \in H^*$ is called the conjugate of T.

An operator T on a Hilbertspace H is said to be an invertible operator if there exists an operator S such that ST=TS=I, where I is the identity operator.

- Special Types of Operators

The special types of operators are defined as follows :

$*$ self-adjoint operator : $T^* = T$

$*$ normal operator : $T^*T = TT^*$

$*$ quasinormal operator : $T(T^*T) = (T^*T)T$

$*$ projection operator : $T^2 = T$ and $T^* = T$

$*$ unitary operator : $T^*T = TT^* = I$

$*$ isometry operator : $T^*T = I$

$*$ positive operator : $\langle Tx, x\rangle \geq 0$ for all $x \in H$

An inner product space X is a vector space with an inner product $\langle x, y\rangle$ defined on it. The latter generalizes the dot product of vectors in three dimensional space and is used to define

- a norm $\|.\|$ by$\langle x\rangle = \langle x, y\rangle^{1/2}$
- orthogonality by$\langle x, y\rangle = 0$

A Hilbert space H is a complete inner product space. The theory of inner product and Hilbert spaces is richer than that of general normed and Banach spaces. The features are $\Rightarrow$ Representation of H as a direct sum of a closed subspace and its orthogonal complement

$\Rightarrow$ Orthonormal sets and sequences and corresponding representations of elements of H

$\Rightarrow$ The Riesz representation of bounded linear functional by inner products

$\Rightarrow$ The Hilbert - adjoint operator T^* of a bounded linear operator T

Orthonormal sets and sequences are truly interesting only if they are total. Hilbert -adjoint operator of a bounded linear operator on a Hilbert space , this operator was suggested ny problems in matrices and linear differential and integral equations. Hilbert -adjoint operators can be used to define classes of operators (self-adjoint, unitary, normal) are of great importance in applications.

SEMI-HILBERTIAN SPACE

In mathematics, a semi hilbert space is a generalization of a hilbert space in functional analysis, in which, roughly speaking, the inner product is required only to be positive semi-definite rather than positive definite , that it gives rise to a seminorm rather than a vector space norm.

The quotient of this space by the kernel of this seminorm is also required to be a Hilbert space in the usual sense.

3.1 A-NORMAL OPERATORS IN SEMI-HILBERTIAN SPACES- 2012

DEFINTION

An operator $T \in \mathcal{L}_A(\mathcal{H})$ is a called an A- normal operator if $T^{\#}T = TT^{\#}$

A-normal operators may be regraded as a generalization of normal and self-adjoint operators in which $T^{\#} = T^{*}$. This last propertyis realized in particular if A=I or if T and A commute and A has a dense range.

The identity operator and the orthogonal projection on $\overline{R(A)}$ are the A- normal.Moreover, if T is an A-normal then$TS, T + S/TS = ST, S = S^{\#}$ is a set of A-normal operators.

Another characterization is that $T \in \mathcal{L}_A(\mathcal{H})$ is an A-normal operator if and only if there are A-selfadjoint operators $B, C \in \mathcal{L}_A(\mathcal{H})$ such that $BC = CB$ and $T = B + iC$, $(i^2 = -1)$.

THEOREM:3.1.1

A necessary and sufficient condtion for an operator$T \in \mathcal{L}_A(\mathcal{H})$ to be A-normal is that $R(TT^{\#}) \subset \overline{R(A)}$ and $\|Tx\|_A = \left\|T^{\#}x\right\|_A$ for every vector $x \in \mathcal{H}$.

PROOF

Suppose that T is A-normal. It is easy that $R(TT^{\#}) = R(T^{\#}T) \subset \overline{R(A)}$. Moreover, using the fact that $TT^{\#}$ is A-selfadjoint, then for $x \in \mathcal{H}$, we obtain

$$\begin{aligned}
T^{\#}T &= TT^{\#} \\
\left\langle T^{\#}Tx, x\right\rangle_A &= \left\langle TT^{\#}x, x\right\rangle_A \\
\left\langle AT^{\#}Tx, x\right\rangle &= \left\langle ATT^{\#}x, x\right\rangle \\
\langle T^*ATx, x\rangle &= \left\langle T(T^{\#})^*Ax, x\right\rangle \\
\langle ATx, Tx\rangle &= \left\langle T^*Ax, T^{\#}x\right\rangle \\
\|Tx\|_A &= \left\|T^{\#}x\right\|_A
\end{aligned}$$

Coversely,if $\|Tx\|_A = \left\|T^{\#}x\right\|_A$,then $A(T^{\#}T - TT^{\#}) = 0$,if moreover $R(TT^{\#}) = R(T^{\#}T) \subset \overline{R(A)}$,so,it follows $R(T^{\#}T - TT^{\#}) \subset \overline{R(A)} = N(A)^{\perp}$ hence $T^{\#}T - TT^{\#} = 0$.

Hence the proof.

3.2 A- QUASI NORMAL OPERATORS IN SEMI-HILBERTIAN SPACES -2012

DEFINTION

An operator$T \in \mathcal{L}_A(\mathcal{H})$ is called A-quasinormal if T commutes with $T^{\#}$ i.e. $T(T^{\#}T) = (T^{\#}T)T$.

Let $T = U + V \in \mathcal{L}_A(\mathcal{H})$ where $U = \frac{T+T^{\#}}{2}$ and $V = \frac{T-T^{\#}}{2}$. We shall write $B^2 = TT^{\#}$ and $C^2 = T^{\#}T$ where B and C are non-negative definite. We give necessary and sufficient condition for an operator to be A-quasinormal.

THEOREM:3.2.1

T is A-quasinormal with N(A) is invariant subspace for T if and only if C commutes with U and V.

PROOF

Since N(A) is invariant subspace for T we observe that $PT = TP$ and $T^{\#}P = PT^{\#}$. Let T be A-quasinormal then

$$\begin{aligned}
T(T^{\#}T) &= (T^{\#}T)T \\
T^{\#}(T^{\#})^{\#}T^{\#} &= T^{\#}T^{\#}(T^{\#})^{\#} \\
T^{\#}PTPT^{\#} &= T^{\#}T^{\#}PTP \\
PT^{\#}PTT^{\#} &= T^{\#}PT^{\#}PT \\
T^{\#}TT^{\#} &= T^{\#}T^{\#}T \\
Hence \quad T^{\#}TT^{\#} &= (T^{\#})^2T
\end{aligned}$$

Now it is easy to se that $C^2U = UC^2$. Since C is non-negative definite, it follows that $CU = UC$. Similarly $CV = VC$.

Coversely,if CU=UC and CV=VC. Then $C^2U = UC^2$ and $C^2v = vC^2$. Hence $C^2T = TC^2$ Therefore, $T^{\#}T^2 = TT^{\#}T$.

Hence the proof.

THEOREM:3.2.2

If T is an operator such that(i) B commutes with U and V (ii) $C^2T = TC^2$. Then T is A-quasinormal.

PROOF

Since $BU = UB$ and $BV = VB$ we have $B^2U = UB^2$ and $B^2V = VB^2$

Then

$$\begin{aligned} B^2T + B^2T^{\#} &= TB^2 + T^{\#}B^2 \\ B^2T - B^2T^{\#} &= TB^2 - T^{\#}B^2 \end{aligned}$$

This gives $B^2T = TB^2 = C^2T$.Hence T is A-quasinormal.

THEOREM:3.2.3

Let T be A-quasi normal $C^2T = TB^2$ and N(A) be an invariant subspace for T. Then B commutes with U and V.

PROOF

Since $C^2T = TB^2$ we have $T^{\#}T^2 = T^2T^{\#}$. Hence $(T^{\#})^2T = T(T^{\#})^2$.

Since T is A-quasi normal we have

$$\begin{aligned} B^2U &= \frac{TT^{\#}T + T(T^{\#})^2}{2} = \frac{T^{\#}T^2 + (T^{\#})^2T}{2} \\ &= \frac{T^2T^{\#} + T^{\#}TT^{\#}}{2} = \frac{T + T^{\#}}{2}TT^{\#} \\ B^2U &= UB^2 \end{aligned}$$

Hence $BU = UB$. Similarly BV=VB.

THEOREM:3.2.4

Let S and T be two A-quasinormal operators. Then their product ST is A-quasinormal if the following conditions are satisfied (i) $ST = TS$ (ii)$ST^{\#} = T^{\#}S$.

PROOF

$$\begin{aligned}
(ST)(ST)^{\#}(ST) &= (ST)(T^{\#}S^{\#})(ST) \\
&= (ST)(S^{\#}T^{\#})(ST) \\
&= S(TS^{\#})(T^{\#}S)T \\
&= SS^{\#}(TS)T^{\#}T \\
&= SS^{\#}(ST)T^{\#}T \\
&= (SS^{\#}S)(TT^{\#}T) \\
&= (S^{\#}S^{2})(T^{\#}T^{2}) \\
&= S^{\#}(S^{2}T^{\#})T^{2} \\
&= S^{\#}(T^{\#}S^{2})T^{2} \\
&= (T^{\#}S^{\#})(S^{2}T^{2}) \\
&= (ST)^{\#}(ST)^{2}
\end{aligned}$$

Hence ST is A-Quasinormal.

THEOREM:3.2.5

Let S and T be two A-quasinormal operators such that $ST = TS = S^{\#}T = T^{\#}S = 0$. Then S+T is A-quasinormal.

PROOF

$$\begin{aligned}
(S+T)(S+T)^{\#}(S+T) &= (S+T)(S^{\#}+T^{\#})(S+T) \\
&= (S+T)(S^{\#}S + S^{\#}T + T^{\#}S + T^{\#}T) \\
&= (S+T)(S^{\#}S + T^{\#}T) \\
&= SS^{\#}S + ST^{\#}T + TS^{\#} + TT^{\#}T \\
&= S^{\#}S^2 + T^{\#}T^2 \\
&= (S+T)^{\#}(S+T)^2
\end{aligned}$$

Hence S+T is A-Quasinormal.

3.3 (α, β) A-NORMAL OPERATORS IN SEMI-HILBERTIAN SPACES -2016

DEFINITION

An operator $T \in \mathcal{B}_A\mathcal{H}$ is a said to be (α, β)-A normal for $0 \leq \alpha \leq 1 \leq \beta$, if

$$\beta^2 T^{\#}T \geq_A TT^{\#} \geq_A \alpha^2 T^{\#}T$$

which is equivalent to the condition

$$\beta \left\|Tu\right\|_A \geq \left\|T^{\#}u\right\|_A \geq \alpha \left\|Tu\right\|_A, forallu \in \mathcal{H}$$

THEOREM:3.3.1

Let $T \in \mathcal{B}_A\mathcal{H}$ and $(\alpha, \beta) \in \Re^2$ such that $0 \leq \alpha \leq 1 \leq \beta$. Then T is (α, β)-A-normal if and only if the following conditions are satisfied

$$\Lambda^2 TT^{\#} + 2\alpha^2\lambda T^{\#}T + TT^{\#} \quad \geq_A 0, forall\lambda \in \Re and \tag{1}$$

$$\Lambda^2 T^{\#}T + 2\lambda TT^{\#} + \beta^4 T^{\#}T \quad \geq_A 0, forall\lambda \in \Re \tag{2}$$

PROOF

Assume that the condtions(1) and (2) are satisfied and prove that T is(α, β)-A-normal.

$$\begin{aligned}
\Lambda^2 TT^{\#} + 2\alpha^2\lambda T^{\#}T + TT^{\#} &\geq_A 0 \\
\left\langle (\Lambda^2 TT^{\#} + 2\alpha^2\lambda T^{\#}T + TT^{\#})u, u \right\rangle &\geq_A 0, \forall u \in \mathcal{H}, \forall\lambda \in \Re \\
\lambda^2 \left\|T^{\#}u\right\|_A^2 + 2\alpha^2\lambda \left\|Tu\right\|_A^2 + \left\|T^{\#}u\right\|_A^2 &\geq_A 0, \forall u \in \mathcal{H}, \forall\lambda \in \Re \\
\alpha \left\|Tu\right\|_A &\leq \left\|T^{\#}u\right\|_A, \forall u \in \mathcal{H}
\end{aligned}$$

Similarly,

$$\begin{aligned}
\Lambda^2 T^{\#}T + 2\lambda TT^{\#} + \beta^4 T^{\#}T &\geq_A 0 \\
\left\langle (\Lambda^2 T^{\#}T + 2\lambda TT^{\#} + \beta^4 T^{\#}T)u, u \right\rangle &\geq_A 0, \forall u \in \mathcal{H}, \forall \lambda \in \Re \\
\lambda^2 \left\| Tu \right\|_A^2 + 2\lambda \left\| T^{\#}u \right\|_A^2 + \beta^4 \left\| Tu \right\|_A^2 &\geq_A 0, \forall u \in \mathcal{H}, \forall \lambda \in \Re \\
\left\| T^{\#}u \right\|_A &\leq \beta \left\| T^{\#}u \right\|_A, \forall u \in \mathcal{H}
\end{aligned}$$

Consequently

$$\alpha \left\| Tu \right\|_A \leq \left\| T^{\#}u \right\|_A \leq \beta \left\| Tu \right\|_A, forallu \in \mathcal{H}$$

So T is (α, β)-A- normal as desired.

Hence the proof

THEOREM:3.3.2

Let $T \in \mathcal{B}_A\mathcal{H}$ such that $\mathcal{N}(A)$ is invariant subspace for T and $0 \leq \alpha \leq 1 \leq \beta$. The following statements hold

1. If T is (α, β)-A- normal, then λ T is a (α, β)-A-normal for $\lambda \in \mathcal{C}$.
2. If T is (α, β)-A- normal, then $T + \lambda$ for $\lambda \in \mathcal{C}$ is a (α, β)-A-normal, if one of the following conditions holds:

- $\mu_A^1(\overline{\lambda}T) \geq 0$
- $\mu_A^1(\overline{\lambda}T) < 0, |\lambda|^2 + 2|\lambda| \left\| T \right\|_A \mu_A^1(\overline{\lambda}T) > 0.$

PROOF

(1) Since $\mathcal{N}(A)$ is invariant subspace for T we observe that $TP_{\overline{R(A)}} = P_{\overline{R(A)}}T$ and $AP_{\overline{R(A)}} = P_{\overline{R(A)}}A = A$. Let T be (α, β)-A-normal then

$$\begin{aligned}
& \beta^2 T^{\#}T \geq_A TT^{\#} \geq_A \alpha^2 T^{\#}T \\
\Leftrightarrow\ & \beta^2 |\lambda|^2 T^{\#}T \geq_A |\lambda|^2 TT^{\#} \geq_A |\lambda|^2 \alpha^2 T^{\#}T \\
\Leftrightarrow\ & A\overline{\lambda}T^{\#}\lambda T \geq A\lambda T\overline{\lambda}T^{\#} \geq \alpha^2 A\overline{\lambda}T^{\#}\lambda T \\
\Leftrightarrow\ & AP_{\overline{R(A)}}\overline{\lambda}T^{\#}\lambda T \geq AP_{\overline{R(A)}}\lambda T\overline{\lambda}T^{\#} \geq \alpha^2 AP_{\overline{R(A)}}\overline{\lambda}T^{\#}\lambda T \\
\Leftrightarrow\ & \beta^2 A(\lambda T)^{\#}(\lambda T) \geq A(\lambda T)(\lambda T)^{\#} \geq \alpha^2 A(\lambda T)^{\#}(\lambda T) \\
\Leftrightarrow\ & \beta^2 (\lambda T)^{\#}(\lambda T) \geq_A (\lambda T)(\lambda T)^{\#} \geq_A \alpha^2 (\lambda T)^{\#}(\lambda T)
\end{aligned}$$

Therefore λT is (α, β)-A-normal operator.

(2) Assume that T is (α, β)-A-normal and the conditon (i) holds. We need to prove that

$$\begin{aligned}
\alpha^2 \left\langle (T+\lambda)^{\#}(T+\lambda)u, u \right\rangle_A &\leq \left\langle (T+\lambda)(T+\lambda)^{\#}u, u \right\rangle_A \\
&\leq \left\langle (T+\lambda)(T+\lambda)^{\#}u, u \right\rangle_A \\
&\leq \beta^2 \left\langle (T+\lambda)^{\#}(T+\lambda)u, u \right\rangle_A
\end{aligned}$$

In order to verify (2)we have

$$
\begin{aligned}
&\alpha^2 \left\langle (T+\lambda)^{\#}(T+\lambda)u, u \right\rangle_A \\
&= \alpha^2 \left\{ \left\langle T^{\#}Tu, u \right\rangle_A + \left\langle \lambda T^{\#}u, u \right\rangle_A + \left\langle \overline{\lambda} P_{\overline{R(A)}} Tu, u \right\rangle_A + |\lambda|^2 \left\langle P_{\overline{R(A)}} Tu, u \right\rangle_A \right\} \\
&= \alpha^2 \left\{ \left\langle T^{\#}Tu, u \right\rangle_A + 2Re \left\langle \overline{\lambda} Tu, u \right\rangle_A + |\lambda|^2 \|u\|_A^2 \right\} \\
&\leq \alpha^2 \left\langle T^{\#}Tu, u \right\rangle_A + \alpha^2 \left\{ 2Re \left\langle \overline{\lambda} Tu, u \right\rangle_A + |\lambda|^2 \|u\|_A^2 \right\}
\end{aligned}
$$

The condition (i) implies that $2Re \left\langle \overline{\lambda} Tu, u \right\rangle_A \geq 0$ and it follows that

$$
\begin{aligned}
\alpha^2 \left\langle (T+\lambda)^{\#}(T+\lambda)u, u \right\rangle_A &\leq \left\{ \left\langle TT^{\#}u, u \right\rangle_A + 2Re \left\langle \overline{\lambda} Tu, u \right\rangle_A + |\lambda|^2 \|u\|_A^2 \right\} \\
&= \left\langle (T+\lambda)(T+\lambda)^{\#}u, u \right\rangle_A \\
&= \left\{ \left\langle TT^{\#}u, u \right\rangle_A + 2Re \left\langle \overline{\lambda} Tu, u \right\rangle_A + |\lambda|^2 \|u\|_A^2 \right\} \\
&\leq \beta^2 \left\langle (T+\lambda)^{\#}(T+\lambda)u, u \right\rangle_A
\end{aligned}
$$

and hence $T+\lambda$ is (α, β)-A-normal. On the other hand if the condition (ii) is statisfied then we have for $\lambda \neq 0$

$$
\begin{aligned}
& |\lambda|^2 + 2|\lambda| \|T\|_A \mu_A^1(\overline{\lambda}T) \\
= \;& |\lambda|^2 + 2|\lambda| \left(\overset{sup}{\|Tu\|_A} \right) \left(inf \left\{ \frac{Re \left\langle \overline{\lambda} Tu, u \right\rangle_A}{|\lambda| \|Tu\|_A}, \|u\|_A = 1, Tu \notin \mathcal{N}(A)^{\frac{1}{2}} \right) \right\} \\
\leq \;& |\lambda|^2 + 2 \overset{inf}{\|u\|_{A=1}} Re \left\langle \overline{\lambda} Tu, u \right\rangle_A \\
\leq \;& |\lambda|^2 + 2Re \left\langle \overline{\lambda} Tu, u \right\rangle_A
\end{aligned}
$$

A similar argument used as above shows that $T+\lambda$ is(α, β)-A-normal.

THEOREM:3.3.3

Let T,V $\in \mathcal{B}_A(\mathcal{H})$ such that $\mathcal{N}(A)$ is invariant subspace for both T and V. If T is an (α, β)-A-normal $(0 \leq \alpha \leq 1 \leq \beta)$ and V is an A-isometry, then $VTV^{\#}$ is an (α, β)-A-normal operator.

PROOF

Assume that $\beta^2 T^{\#}T \geq_A TT^{\#} \geq_A \alpha^2 T^{\#}V$ and $V^{\#}V = P_{\overline{R(A)}}$. This implies

$$\begin{aligned}
\beta^2(VTV^{\#})^{\#}(VTV^{\#}) &= \beta((V^{\#})^{\#}T^{\#}V^{\#}VTV^{\#}) \\
&= \beta^2(P_{\overline{R(A)}}VP_{\overline{R(A)}}T^{\#}P_{\overline{R(A)}}TV^{\#}) \\
&= \beta^2(VP_{\overline{R(A)}}T^{\#}T(VP_{\overline{R(A)}})^{\#}) \\
&\geq_A VP_{\overline{R(A)}}TT^{\#}(VP_{\overline{R(A)}})^{\#} \\
&\geq_A (VTV^{\#})(VTV^{\#})^{\#}
\end{aligned}$$

Similarly, we have

$$\begin{aligned}
(VTV^{\#})(VTV^{\#})^{\#} &= VP_{\overline{R(A)}}TT^{\#}(VP_{\overline{R(A)}})^{\#} \\
&\geq_A \alpha^2(VP_{\overline{R(A)}}T^{\#}T(VP_{\overline{R(A)}})^{\#}) \\
&\geq_A \alpha^2(VTV^{\#})^{\#}(VTV^{\#})
\end{aligned}$$

Hence the theorem holds.

THEOREM:3.3.4

Let T,S $\in \mathcal{B}_A(\mathcal{H})$ such that T is (α, β)-A-normal and S is A- selfadjoint. If $T^{\#}S = ST^{\#}$ then TS is (α, β)-A-normal.

PROOF

Since T is (α, β)-A-normal we have for u $\in \mathcal{H}$

$$\alpha \left\| TSu \right\|_A \leq \left\| T^{\#}Su \right\|_A \leq \beta \left\| TSu \right\|_A$$

On the other hand

$$\begin{aligned} \left\| T^{\#}Su \right\|_A^2 &= \left\langle T^{\#}Su, T^{\#}Su \right\rangle_A \\ &= \left\langle AST^{\#}u, ST^{\#}u \right\rangle \\ &= \left\langle (TS)^{\#}u, (TS)^{\#}u \right\rangle_A \\ &= \left\| (TS)^{\#}u \right\|_A^2 \end{aligned}$$

This implies

$$\alpha \left\| TSu \right\|_A \leq \left\| T^{\#}Su \right\|_A \leq \beta \left\| TSu \right\|_A$$

THEOREM:3.3.5

Let T,V $\in \mathcal{B}_A(\mathcal{H})$ such that T is (α, β)-A-normal and S is A-unitary. If TS=ST and $\mathcal{N}(A)$ is invariant subspace for T then TS is (α, β)-A-normal.

PROOF

Since $\mathcal{N}(A)$ is invariant subspace for T we observe $TP_{\overline{R(A)}} = P_{\overline{R(A)}}T$ and $T^{\#}P_{\overline{R(A)}} = P_{\overline{R(A)}}T^{\#}$. Let S be A-unitary then $S^{\#}S = SS^{\#} = P_{\overline{R(A)}}$.

Now it is easy to see that

$$\begin{aligned}\beta^2((TS)^{\#}(TS)) &= \beta^2(T^{\#}S^{\#}ST)\\ &= \beta^2(T^{\#}P_{\overline{R(A)}}T)\\ &= \beta^2(P_{\overline{R(A)}}T^{\#}TP_{\overline{R(A)}})\end{aligned}$$

By using the fact that T is (α, β)-A-normal,

$$\beta^2((TS)^{\#}(TS)) \geq_A \underbrace{(P_{\overline{R(A)}}TT^{\#})}_{1} \geq_A \alpha^2 \underbrace{P_{\overline{R(A)}}T^{\#}T}_{2}$$

Notice that (1) gives

$$\begin{aligned}P_{\overline{R(A)}}TT^{\#}P_{\overline{R(A)}} &= TP_{\overline{R(A)}}T^{\#}\\ &= TSS^{\#}T^{\#}\\ &= TS(TS)^{\#}\end{aligned}$$

and similarly (2) gives

$$\begin{aligned}P_{\overline{R(A)}}T^{\#}TP_{\overline{R(A)}} &= T^{\#}P_{\overline{R(A)}}T\\ &= T^{\#}S^{\#}ST\\ &= (TS)^{\#}(TS)\end{aligned}$$

So $\beta^2(TS)^{\#}(TS) \geq_A TS(TS)^{\#} \geq_A \alpha^2(TS)^{\#}(TS)$

Hence TS is(α, β)-A-normal operator.

THEOREM:3.3.6

Let T,S $\in \mathcal{B}_A(\mathcal{H})$ such that T is (α, β)-A-normaL $(0 \geq \alpha \geq 1 \geq \beta)$ and S is (α', β') A-normal $(0 \geq \alpha' \geq 1 \geq \beta')$. Then the foollowing statements hold:

1. If $T^{\#}S = ST^{\#}$, then TS is $(\alpha\alpha', \beta\beta')$-A-normal operator.
2. If $S^{\#}T = TS^{\#}$, then ST is $(\alpha\alpha', \beta\beta')$-A-normal operator.

PROOF

(1) Since T is (α, β)-A-normal and S is (α', β')-A-normal,it follows that for all $u \in \mathcal{H}$

$$\begin{aligned}
\alpha\alpha' \|TSu\|_A \leq \alpha' \left\|T^{\#}Su\right\|_A &= \alpha' \left\|ST^{\#}u\right\|_A \\
&\leq \left\|S^{\#}T^{\#}u\right\|_A \\
&= \left\|(TS)^{\#}u\right\|_A \\
&= \beta' \left\|T^{\#}Su\right\|_A \\
&\leq \beta\beta' \|TSu\|_A
\end{aligned}$$

It follows that

$$\alpha\alpha' \|STu\|_A \leq \left\|(TS)^{\#}u\right\|_A \leq \beta\beta' \|TSu\|_A$$

The proof of the second assertion is completed in much the same way as the first assertion.

3.4 (n,m)-POWER QUASI NORMAL OPERATORS IN SEMI-HILBERTIAN SPACES -2017

DEFINITION

An operator $T \in \mathcal{B}_A(\mathcal{H})$ is said to be be (n,m)-quasinormal operators in for a positive integer (n,m) if

$$T^n(T^{\#})^m T = (T^{\#})^m T T^n$$

THEOREM:3.4.1

Let $T \in \mathcal{B}_A(\mathcal{H})$,$B = T^n + T^{\#m}T$ and $C = T^n - T^{\#m}T$. Then we have T$\in$ A-(n,m)-quasinormal if and only if B commutes with C.

PROOF

TO PROVE: $T^n(T^{\#})^m T = (T^{\#})^m T T^n$ BC=CB

Consider,

$$\begin{aligned} BC &= (T^n + T^{\#m}T)(T^n - T^{\#m}T) \\ &= T^n T^n - T^{n+1}(T^{\#m} + (T^{\#m}T^{n+1})^k T^m - (T^{\#m}T)(T^{\#m}T) \end{aligned}$$

Similarly,consider

$$\begin{aligned} CB &= (T^n - T^{\#m}T)(T^n + T^{\#m}T) \\ &= T^nT^n + T^{n+1}(T^{\#m} - (T^{\#m}T^{n+1})^kT^m - (T^{\#m}T)(T^{\#m}T) \end{aligned}$$

Therefore, B commutes with C then T$\in$ A-(n,m,k)-quasinormal

THEOREM:3.4.2

Let S,T$\in$ A-(n,m)-quasinormal such that $ST = TS = S^{\#m}T = T^{\#m}S = 0$. Then S+T is A-(n,m)-quasinormal.

PROOF

TO PROVE: $T^n(T^{\#})^mT = (T^{\#})^mTT^n$
$(S+T)^n((S+T)^{\#m}(S+T)) = ((S+T)^{\#m}(S+T))(S+T)^n$

Consider,

$$\begin{aligned} (S+T)^n((S+T)^{\#m}(S+T)) &\Leftrightarrow (S^n + T^n)((S^{\#m} + T^{\#m})(S+T)) \\ &\Leftrightarrow (S^n + T^n)(S^{\#m}S + S^{\#m}T + T^{\#m}S + T^{\#m}T) \\ &\Leftrightarrow (S^n + T^n)((S^{\#m}S)^+(T^{\#m}T)) \\ &\Leftrightarrow (S^n(S^{\#m}S) + S^n(T^{\#m}T) + T^n(S^{\#m}S) + T^n(T^{\#m}T)) \\ &\Leftrightarrow ((S^{\#m}S) + (T^{\#m}T))(S^n + T^n) \\ &\Leftrightarrow ((S+T)^{\#m}(S+T))(S+T)^n \end{aligned}$$

Therefore, S+T is A-(n,m,k)-quasinormal.

THEOREM:3.4.3

Let $S, T \in A{-}(n,m)$-quasinormal such that $ST = TS, S^{\#m}T = T^{\#m}S, T^{\#m}S = ST^{\#m}$. Then TS is $A-(n,m)$-quasinormal.

PROOF

TO PROVE: $T^n(T^{\#})^m T = (T^{\#})^m T T^n$
$(TS)^n((TS)^{\#m}(TS)) = ((TS)^{\#m}(TS))(TS)^n$

Consider,

$$\begin{aligned}
((TS)^n((TS)^{\#m}(TS)) &\Leftrightarrow (T^nS^n)((T^{\#m}S^{\#m})(TS)) \\
&\Leftrightarrow (T^nS^n)(T^{\#m}S^{\#m}(TS)) \\
&\Leftrightarrow (T^n(T^{\#m})S^nT(S^{\#m})S) \\
&\Leftrightarrow (T^n(T^{\#m})S^n(S^{\#m})S) \\
&\Leftrightarrow (T^{\#m}T)T^n.(S^{\#m}S)S^n \\
&\Leftrightarrow (T^{\#m})(T)T^n(S^{\#m})(S)S^n \\
&\Leftrightarrow (T^{\#m})(S^{\#m})(T)(S)T^nS^n \\
&\Leftrightarrow ((T^{\#m}S^{\#m})(TS)(T^nS^n) \\
&\Leftrightarrow ((TS)^{\#m}(TS))(TS)^n \\
&\Leftrightarrow ((TS)^{\#m}(TS))(TS)^n
\end{aligned}$$

Therefore, TS is A-(n,m,k)-quasinormal.

(n,m,k)-POWER QUASI NORMAL OPERATORS IN SEMI-HILBERTIAN SPACES

DEFINITION

An operator $T \in \mathcal{B}_A(\mathcal{H})$ is said to be be (n,m,k)-quasinormal operators in for a positive integer n,m,k if

$$T^m(T^{\#}T^n)^k = (T^{\#}T^n)^k T^m$$

THEOREM:4.1

Let $T \in \mathcal{B}_A(\mathcal{H})$,$B = T^m + (T^{\#}T^n)^k$ and$C = T^m - (T^{\#}T^n)^k$. Then we have T$\in$ A-(n,m,k)-quasinormal if and only if B commutes with C.

PROOF

TO PROVE: $T^m(T^{\#}T^n)^k = (T^{\#}T^n)^k T^m$
BC=CB

Consider,

$$\begin{aligned} BC &= (T^m + (T^{\#}T^n)^k)(T^m - (T^{\#}T^n)^k) \\ &= T^mT^m - T^m(T^{\#}T^n)^k + (T^{\#}T^n)^k T^m - (T^{\#}T^n)^k(T^{\#}T^n)^k \end{aligned}$$

Similarly, consider

$$\begin{aligned} CB &= (T^m - (T^{\#}T^n)^k)(T^m + (T^{\#}T^n)^k) \\ &= T^mT^m + T^m(T^{\#}T^n)^k - (T^{\#}T^n)^kT^m - (T^{\#}T^n)^k(T^{\#}T^n)^k \end{aligned}$$

Therefore, B commutes with C then T$\in$ A-(n,m,k)-quasinormal

THEOREM: 4.2

Let S,T$\in$ A-(n,m,k)-quasinormal such that $ST = TS = S^{\#}T^{\#} = T^{\#}S^{\#} = 0$. Then S+T is A-(n,m,k)-quasinormal.

PROOF

TO PROVE: $T^m(T^{\#}T^n)^k = (T^{\#}T^n)^kT^m$
$(S+T)^m((S+T)^{\#}(S+T)^n)^k = ((S+T)^{\#}(S+T)^n)^k(S+T)^m$

Consider

$$\begin{aligned} (S+T)^m((S+T)^{\#}(S+T)^n)^k &\Leftrightarrow (S^m + T^m)((S^{\#} + T^{\#})(S^n + T^n))^k \\ &\Leftrightarrow (S^m + T^m)(S^{\#}S^n + S^{\#}T^n + T^{\#}S^n + T^{\#}T^n)^k \\ &\Leftrightarrow (S^m + T^m)((S^{\#}S^n)^k + (T^{\#}T^n)^k) \\ &\Leftrightarrow (S^m(S^{\#}S^n)^k + S^m(T^{\#}T^n)^k + T^m(S^{\#}S^n)^k + T^m(T^{\#}T^n)^k) \\ &\Leftrightarrow ((S^{\#}S^n)^k + (T^{\#}T^n)^k)(S^m + T^m) \\ &\Leftrightarrow ((S+T)^{\#}(S+T)^n)^k(S+T)^m \end{aligned}$$

Therefore, S+T is A-(n,m,k)-quasinormal.

THEOREM: 4.3

Let S,T$\in$ A-(n,m,k)-quasinormal such that $ST = TS = S^{\#}T^{\#} = T^{\#}S^{\#} = 0$. Then T-S is A-(n,m,k)-quasinormal.

PROOF

TO PROVE: $T^m(T^{\#}T^n)^k = (T^{\#}T^n)^kT^m$
$(T-S)^m((T-S)^{\#}(T-S)^n)^k = ((T-S)^{\#}(T-S)^n)^k(T-S)^m$

Consider

$$\begin{aligned}
(T-S)^m((T-S)^{\#}(T-S)^n)^k &\Leftrightarrow (T^m - S^m)((T^{\#} - S^{\#})(T^n - S^n))^k \\
&\Leftrightarrow (T^m - S^m)((T^{\#}T^n - T^{\#}S^n - S^{\#}T^n + S^{\#}S^n))^k \\
&\Leftrightarrow (T^m - S^m)((T^{\#}T^n)^k + (S^{\#}S^n)^k) \\
&\Leftrightarrow (T^m(T^{\#}T^n)^k + T^m(S^{\#}S^n)^k - S^m(T^{\#}T^n)^k + S^m(S^{\#}S^n)^k) \\
&\Leftrightarrow ((T^{\#}T^n)^k - (S^{\#}S^n)^k)(T^m - S^m) \\
&\Leftrightarrow ((T-S)^{\#}(T-S)^n)^k(T-S)^m
\end{aligned}$$

Hence, $T - SisA - (n, m, k)$-quasinormal.

THEOREM: 4.4

Let S,T$\in$ A-(n,m,k)-quasinormal such that ST=TS, $S^{\#}T = T^{\#}S$, $T^{\#}S = ST^{\#}$. Then TS is A-(n,m,k)-quasinormal.

PROOF

TO PROVE:$T^m(T^{\#}T^n)^k = (T^{\#}T^n)^kT^m$
$(TS)^m((TS)^{\#}(TS)^n)^k = ((TS)^{\#}(TS)^n)^k(TS)^m$

L.H.S

$$\begin{aligned}
(TS)^m((TS)^{\#}(TS)^n)^k &\Leftrightarrow (T^mS^m)((T^{\#}S^{\#})(T^nS^n))^k \\
&\Leftrightarrow (T^mS^m)((T^{\#})^k(S^{\#})^k(T^n)^k(S^n)^k) \\
&\Leftrightarrow (T^m(T^{\#})^kS^m(T^n)^k(S^{\#})^k(S^n)^k) \\
&\Leftrightarrow (T^m(T^{\#}T^n)^kS^m(S^{\#}S^n)^k \\
&\Leftrightarrow (T^{\#}T^n)^kT^m.(S^{\#}S^n)^kS^m \\
&\Leftrightarrow (T^{\#})^k(T^n)^kT^m(S^{\#})^k(S^n)^kS^m \\
&\Leftrightarrow (T^{\#})^k(S^{\#})^k(T^n)^k(S^n)^kT^mS^m \\
&\Leftrightarrow (T^{\#})^k(S^{\#})^k(T^n)^k(S^n)^k)(T^mS^m) \\
&\Leftrightarrow ((T^{\#}S^{\#})(T^nS^n))^k(T^mS^m) \\
&\Leftrightarrow ((TS)^{\#}(TS)^n)^k(TS)^m
\end{aligned}$$

Therefore, TS is A-(n,m,k)-quasinormal.

THEOREM: 4.5

If T is $A-(n,m,k)-quasinormal$ then T^P is also an $A-(n,m,k)-quasinormal$ for P natural numbers.

PROOF

The result is true for P=1

That is,

$$T^m(T^{\#}T^n)^k = (T^{\#}T^n)^k T^m \tag{3}$$

By using induction method, assume that the result for P=S

$$(T^m(T^{\#}T^n)^k)^S = ((T^{\#}T^n)^k T^m)^S \tag{4}$$

Let us prove the result for P=S+1

$$(T^m(T^{\#}T^n)^k)^{S+1} = ((T^{\#}T^n)^k T^m)^{S+1}$$

$$(T^m(T^{\#}T^n)^k)^{S+1} \quad = (T^m(T^{\#}T^n)^k)^S T^m (T^{\#}T^n)^k$$

By (3) and (4)

$$\Leftrightarrow ((T^{\#}T^{n})^{k}T^{m})^{S}(T^{\#}T^{n})^{k}T^{m}$$

$$\Leftrightarrow ((T^{\#}T^{n})^{k}T^{m})^{S+1}$$

$$Therefore \quad (T^{m}(T^{\#}T^{n})^{k})^{S+1} = ((T^{\#}T^{n})^{k}T^{m})^{S+1}$$

Hence T^{P} is also an $A-(n,m,k)-quasinormal$ for P natural numbers

CONCLUSION

In this project,I have introduced a new class of operator (n,m,k)-Power Quasinormal operators on Semi-Hilbertian spaces which is an extension of (n,m)-Power Quasinormal operators . I have analyzed commutativity, sum, subtraction, product and powers of (n,m,k)-Power Quasinormal operators on Semi-Hilbertian spaces. In further studies this work will be extended and serves as a tool for other works.

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